

Wireless Transmission Performance of LoRa with Embedded Lightweight Encryption Schemes

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Abstract—LoRa is a low-power radio frequency-based wireless communication technology capable of transmitting data over long distances. This technology is widely used in embedded systems and the Internet of Things (IoT) in various sectors such as agriculture, fisheries, industry, transportation, and smart cities. However, data transmitted over the LoRa network is not encrypted by default, so the confidentiality of the data is not guaranteed because it can be intercepted and read by other parties at the same frequency. Therefore, data encryption techniques are needed in LoRa systems to maintain data confidentiality and security during the transmission process. This paper presents a comparative study of three lightweight encryption algorithms PRESENT, CHACHA20, and SPECK implemented on LoRa-based systems using RFM95W modules and Arduino R4 WiFi microcontrollers. Encrypted messages are transmitted between two LoRa nodes at varying distances to observe the effect of encryption on the quality of wireless communication. Performance evaluation is carried out based on three main metrics, namely signal strength (RSSI), communication delay time (latency), and transmission range. The test results show that each encryption algorithm has a different impact on LoRa communication performance, particularly in terms of the efficiency and reliability of long-distance data transmission. These findings provide practical insights for developers and practitioners who need secure yet responsive and stable LoRa communication.

Index Terms—LoRa, lightweight encryption, IoT, network performance

I. INTRODUCTION

LoRa (Long Range) technology has been developed to support communication needs in Internet of Things (IoT) systems, offering low power consumption and long-range coverage [1]. One standout LoRa technology, known for its cost efficiency and flexibility, IoT devices are rapidly

expanding into various fields such as home automation, healthcare, and agriculture [2].

LoRa is a radio frequency-based communication protocol designed for low power consumption and long-distance transmission, making it ideal for supporting the implementation of Internet of Things (IoT) systems in remote areas with limited infrastructure, especially in the smart agriculture sector [3]. Beyond agriculture, LoRa technology is also being adopted in environmental monitoring applications, such as early forest fire detection and water resource management in remote areas that lack conventional communication infrastructure [4].

Despite its advantages, the growing use of LoRa in real-world applications raises critical security concerns [5]. The protocol's security largely depends on how it is implemented by developers or device manufacturers [6]. While its wide coverage with minimal infrastructure makes it ideal for smart cities and real-time disaster management systems [7], the success of such applications relies heavily on the reliability and security of data transmission between IoT devices within the network [8].

Although LoRa offers efficiency and flexibility, its security largely depends on how it is implemented by developers or device manufacturers. Improper implementation can create vulnerabilities that may be exploited by third parties [9]. This threat is becoming increasingly relevant as the network is integrated into critical systems and sensitive data is collected in real-world environments. Without proper authentication and encryption mechanisms, the communications can be exploited by unauthorized parties through attacks such as sniffing, replay attacks, injection, and even jamming that are predicted based on airtime

and payload patterns [10]. Therefore, lightweight and efficient encryption mechanisms are required to ensure data confidentiality and integrity without compromising communication performance [11].

Lightweight encryption has emerged as an ideal solution for IoT devices with limited resources, such as low-power processors and restricted memory capacity. In this context, the PRESENT and SPECK encryption algorithms have demonstrated efficient use of memory and energy, making them well-suited for resource-constrained embedded devices [12]. Additionally, ChaCha20 is also a strong candidate due to its high efficiency and strong security for IoT applications [13].

LoRa-based devices are especially affected by these trade-offs, as heavier encryption mechanisms can increase power consumption, latency, and message size, potentially hindering communication efficiency [14], [15]. Furthermore, the cipher-text size resulting from encryption can also affect LoRa packet transmission, which has a maximum payload limit. Therefore, choosing an appropriate algorithm requires careful consideration of transmission overhead [16]. It is thus essential to strike a balance between strong security and efficient resource usage, such as cost and performance, when selecting encryption algorithms for constrained IoT devices [17].

This study aims to compare the performance of the PRESENT, ChaCha20, and SPECK encryption algorithms within a LoRa communication system implemented using the RFM95W module and the Arduino R4 WiFi microcontroller. The evaluation is based on three key metrics: *received signal strength indicator* (RSSI), *latency*, and *transmission range*. By comparing these three algorithms, the study aims to provide recommendations on the optimal encryption algorithm to ensure data security while maintaining reliable wireless communication in LoRa networks.

II. RELATED WORKS

The use of LoRa (Long Range) in IoT communication systems is growing rapidly in line with the increasing need for power-efficient wireless solutions with wide coverage. However, communication security is a particular concern. In 2020, research indicated that the LoRaWAN protocol is vulnerable to various attacks such as spoofing and denial of service (DoS) if not accompanied by additional security mechanisms, emphasizing the importance of integrating cryptography into LoRa systems [18].

Moving into 2021, there was research developing a real-time health monitoring system based on edge-fog-cloud architecture that emphasised energy efficiency [19]. This study highlights the importance of energy efficiency in medical data processing, but does not specifically address data quality parameters such as RSSI or latency.

More applied studies began to emerge in the context of LoRa, implementing the AES-256 algorithm in a LoRa-based asset tracking system, which ensures a high

level of security but has not yet assessed its impact on communication metrics such as RSSI and latency [20]. There is also research proposing SPECK as a lightweight encryption algorithm suitable for edge-IoT systems in the healthcare field due to its efficiency in memory usage and processing speed, though it has not been directly tested on LoRa communication systems [21]. Conversely, there was an experimental evaluation of RSSI and SNR values using a machine clustering approach in the LoRa-ESL system, but without direct relevance to the encryption algorithm [22].

Moving into 2023, there is research offering a different approach by integrating hardware cryptographic accelerators into the LoRa system. The goal is to accelerate the encryption process without compromising communication performance, which is highly relevant for edge devices with resource constraints [23]. Meanwhile, another study evaluates the performance of several lightweight cryptographic algorithms such as PRESENT and SPECK on Arduino and Raspberry Pi microcontroller platforms. This study emphasizes the importance of balancing security and efficiency, although it does not explicitly test their impact on LoRa transmission [24].

Entering 2024, there is research using RSSI and SNR values in training machine learning models for LoRa-based indoor localization systems. This research confirms that signal quality is an important factor in system accuracy, although it does not link the influence of encryption algorithms on communication performance [25].

A recent study by Wang et al. (2025) developed a LoRa-based electrical energy monitoring system for low-voltage users. This study focused on long-term data transmission efficiency and supports the need for low-power wireless communication for smart city applications, although it did not explicitly investigate lightweight encryption aspects [26].

From this review, it is evident that most previous studies have focused solely on the efficiency of encryption algorithms or communication quality separately. No study has specifically compared the direct impact of lightweight encryption algorithms on signal quality (RSSI) and latency in LoRa communication systems. Therefore, this study aims to address this gap by testing and analyzing the performance of three lightweight encryption algorithms—PRESENT, ChaCha20, and SPECK—in terms of communication quality and speed in low-power microcontroller-based LoRa systems.

III. METHODOLOGY

A. System Design

The communication system in this study was built using two LoRa nodes, each consisting of an Arduino R4 WiFi microcontroller and a LoRa RFM95W module. One node is configured as the transmitter, while the other node serves as the receiver. Both transmitter and receiver nodes

are equipped with OLED screens that display encrypted and decoded messages, respectively. The hardware design circuit used in this system is shown in Fig 1, which illustrates the relationship between major components such as the microcontroller, LoRa module, and OLED display. All devices are powered via a USB connection from a smartphone or power bank, making the system suitable for field testing.

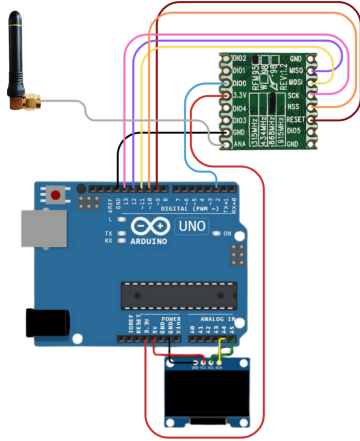


Fig. 1. Hardware Circuit Design.

The program on each microcontroller is developed using the Arduino IDE in C++. The sending node encrypts the message in the string format “message (number)” using one of three lightweight encryption algorithms-PRESENT, ChaCha20, or SPECK-before sending it over the LoRa network at a frequency of 915 MHz. This process is described in Fig 2 which displays a flowchart of the LoRa Transmitter program implementation, from message generation, encryption process, to data transmission.

The receiver node captures the encrypted text, decrypts it using the appropriate algorithm, and displays the resulting original text on the OLED screen. The complete workflow on the receiver side can be seen in Fig 3, which shows a flowchart of the LoRa Receiver program implementation design. The system is designed to be efficient, lightweight, and in keeping with the characteristics of real-world IoT devices that operate with limited resources.

B. Encryption Algorithm Implementation

This study uses three lightweight encryption algorithms commonly applied in embedded systems and the Internet of Things (IoT), namely PRESENT, CHACHA20, and SPECK. These three were chosen because they have low complexity, small memory requirements, and high efficiency on devices with limited resources such as Arduino microcontrollers. PRESENT is a block cipher algorithm based on a Substitution-Permutation Network (SPN) with a block size of 64 bits and a key size of 80/128 bits, consisting of 31 rounds. This algorithm is known for its very low resource consumption, making it suitable for

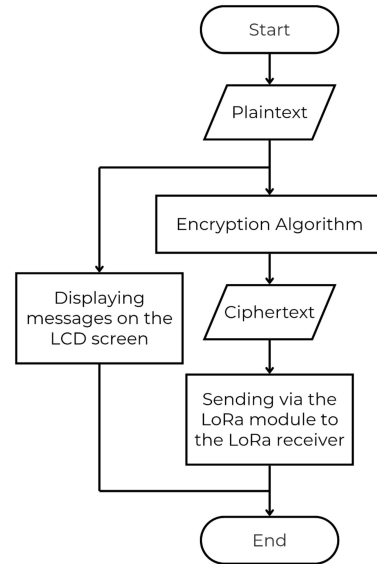


Fig. 2. Flowchart Transmitter.

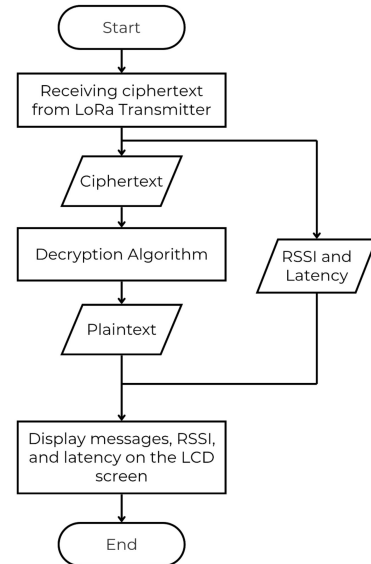


Fig. 3. Flowchart Receiver.

implementation on IoT devices. Meanwhile, CHACHA20 is a modern stream cipher algorithm with a 256-bit key and a 96-bit nonce that offers high security and stable encryption speed.

In this study, the three algorithms were implemented on an Arduino UNO R4 WiFi platform connected to an RFM95W LoRa module. The encryption process was performed on the sender side before the data was transmitted via LoRa, and the decryption process was performed on the receiver side after the data was received. All algorithms were tested using the same parameters and data transmission scenarios, enabling a fair performance evaluation based on the defined metrics.

The following are the results of the input and output of each encryption algorithm. Based on Table I, the encryption process in the three algorithms uses a 64-bit data block with different key sizes, namely 80 bits for PRESENT, 96 bits for SPECK, and 256 bits for CHACHA20. The number of encryption rounds used also varies, with 31 rounds for PRESENT, 26 rounds for SPECK, and 20 rounds for CHACHA20. The total message length (sensor data) used in this test is 64 bytes, which is divided and encrypted in blocks according to each algorithm's specifications. The table presents the encryption results in the form of cipher-text from each algorithm for the same plain text input, as a basis for verifying that each implementation has been executed correctly before performance testing such as signal strength, encryption time, and power consumption.

C. Testing Scheme

System testing was conducted in an open area with minimal physical interference and electromagnetic interference. The purpose of this testing was to evaluate the effect of encryption algorithms on the quality of wireless communication using LoRa. The transmitter node continuously sends encrypted messages in the format "message (number)", with the number increasing sequentially throughout the testing and not returning to the initial number. This pattern is used so that each message has a unique identity, making it easier to ensure the order of reception and detect if any messages are lost or delayed.

The testing was conducted in stages from 10 meters to 100 meters, focusing on two main parameters: RSSI to measure received signal strength and latency calculated based on the time difference between message transmission and reception. Each encryption algorithm was retested using the same scenario and parameters, ensuring each received equal treatment and allowing results to be compared objectively.

IV. RESULT AND DISCUSSION

The testing was conducted by sending data between two LoRa nodes (Transmitter and Receiver) using three encryption algorithms: PRESENT, ChaCha20, and SPECK. Each algorithm was tested with a data transmission scenario at a fixed distance of 100 metres in an open environment with a distance of 10 metres between each node. Information on the test location is shown in Fig 4.

The testing focused on two main parameters, namely *received signal strength indicator* (RSSI) and *latency* or transmission delay time. RSSI measurements were taken to assess the strength of the data reception signal between two LoRa nodes. The testing was conducted on the LoRa system by comparing conditions without encryption and with the application of three encryption algorithms, namely ChaCha20, PRESENT, and SPECK. The aim was to evaluate and compare the performance of each in terms of signal strength (RSSI). The test results are displayed

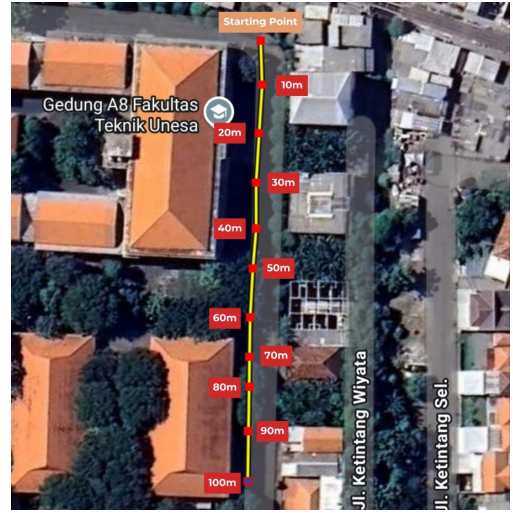


Fig. 4. Data Communication Test Location Map.

together in a single graph in Fig 5, which shows the comparison of RSSI values from the four conditions at various test distances.

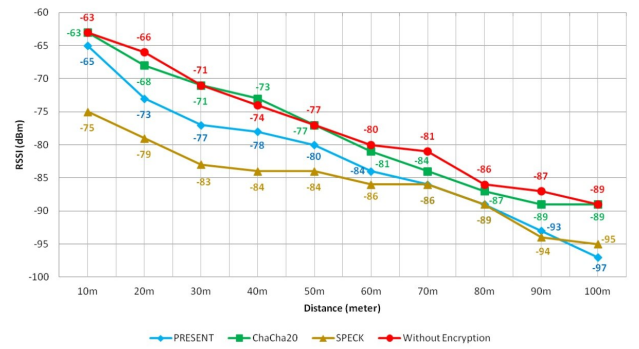


Fig. 5. Comparison of RSSI Values.

Based on the results shown, all scenarios indicate a decrease in signal strength as distance increases, which is a common characteristic of wireless communication. However, there are significant differences in performance between algorithms. As shown in Fig 5, the ChaCha20 algorithm consistently records the highest RSSI values at every test distance compared to other algorithms. For instance, at a distance of 10 meters, the RSSI value reaches approximately -63 dBm and remains stable at around -89 dBm at 100 meters. This suggests that ChaCha20 demonstrates the best and most consistent RSSI performance among the tested encryption methods.

This superior performance can be attributed to the fact that ChaCha20 is a stream cipher, which typically processes data in smaller, continuous units without significantly impacting the timing and modulation characteristics of the signal. As a result, the influence of encryption on signal strength is minimal. Moreover, the RSSI values of ChaCha20 are close to those recorded under unencrypted

TABLE I
ENCRYPTION ALGORITHM TEST DATA

PRESENT Algorithm		
Plaintext	Key	Ciphertext
706573616E203000	EEFFDDCCBBAA 99887766	BEBD42D8BDB74474
706573616E203100	EEFFDDCCBBAA 99887766	2CE7E7C43D341298
706573616E203200	EEFFDDCCBBAA 99887766	45086E6C4D8BBBEF
ChaCha20 Algorithm		
Plaintext	Key	Ciphertext
706573616E203000	00010203040506 0708090A0B0C0D0E0F 1011121314151617 18191A1B1C1D1E1F	FAB9E29CF1D4C0F5
706573616E203100	00010203040506 0708090A0B0C0D0E0F 1011121314151617 18191A1B1C1D1E1F	60949485BF1B6815
706573616E203200	00010203040506 0708090A0B0C0D0E0F 1011121314151617 18191A1B1C1D1E1F	7AEDF01657F78D4E
SPECK Algorithm		
Plaintext	Key	Ciphertext
706573616E203000	332211007766 5544BBAA9988	EDE74A9DE2A07365
706573616E203100	332211007766 5544BBAA9988	A7C42736F5DDFF6D
706573616E203200	332211007766 5544BBAA9988	C006138A14F5B2DA

conditions, reinforcing the observation that ChaCha20 has minimal negative impact on signal degradation.

The results of this experiment indicate that the use of ChaCha20, despite its relatively high computational complexity, has the least impact on signal strength degradation in its communications. The consistently strong RSSI values indicate that this algorithm can effectively preserve signal integrity over various distances. This characteristic is particularly valuable in scenarios requiring both robust encryption and stable transmission quality.

In addition to signal strength, latency is also an important consideration. Similar to RSSI testing, latency parameter testing was also conducted by comparing conditions without encryption and with the application of three encryption algorithms, namely ChaCha20, PRESENT, and SPECK, to assess the extent to which the application of encryption algorithms affects latency in LoRa communication. The test results are displayed together in a single graph in Fig 6, showing the comparison of latency values from the four conditions at various test distances.

Based on the graph, the ChaCha20 algorithm consistently shows the highest latency at all distances, ranging from 2149 to 2151 ms. This reveals that although ChaCha20 excels in terms of signal strength (RSSI), its computational complexity results in a significant increase in latency. This underscores the trade-off between security strength and processing efficiency in resource-constrained IoT systems.

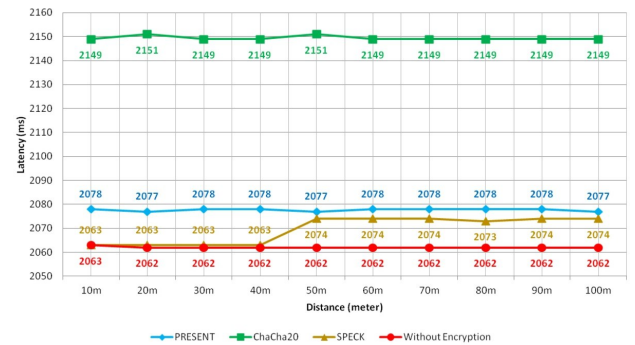


Fig. 6. Comparison of Latency Values.

Conversely, the results show that not all encryption algorithms lead to significant latency increases. The SPECK algorithm demonstrates latency performance closest to the unencrypted scenario, with stable and relatively low values across all tested distances. This reflects high computational efficiency and a low processing load, resulting in response times nearly equivalent to those of unencrypted systems. These findings highlight the importance of selecting encryption schemes not only based on their security strength but also considering real-time communication needs and energy efficiency in IoT environments.

V. CONCLUSION

This study evaluates the performance of three lightweight encryption algorithms, namely ChaCha20, PRESENT, and SPECK, on LoRa communication systems, focusing on signal strength (RSSI) and latency parameters. The test results show that all algorithms experience a decrease in RSSI as the distance increases, but ChaCha20 provides the highest RSSI value and is closest to the unencrypted condition. This indicates that ChaCha20 does not significantly affect the signal strength.

However, in terms of latency, ChaCha20 produces the highest delay due to its encryption complexity, while SPECK shows the lowest latency and is closest to the no-encryption system. This reflects the processing efficiency of the SPECK algorithm without burdening the network capacity.

Thus, the selection of encryption algorithms in LoRa should be tailored to the needs of the system. ChaCha20 is more suitable for applications that prioritize signal stability and security, while SPECK is more ideal for applications with fast response time demands. These findings are expected to be a reference in the development of a secure and efficient IoT communication system.

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